

Lecture 14

很多时候我们需要检验多个 hypotheses :

- ① pairwise multiple hypothesis testing (逐个检验多个 hypotheses 是否成立)
e.g. 判断病人与正常人之间, 具体哪些基因发生了差异表达.
- ② global testing / simultaneous multiple hypothesis testing (检验多个 hypotheses 是否同时成立)
e.g. 判断病人与正常人之间, 是否有基因发生了差异表达.

假设 significance level 为 α , 共有 m 个 (独立的) subhypotheses.

倘若对每个 subhypothesis 都进行 size α test, 那么整体来说, 犯 type-I error 的概率是 $1 - (1 - \alpha)^m \rightarrow 1$ as $m \rightarrow \infty$. 因此我们需要做出改进, 主要有以下思路:

- ① 控制 FWER (Family-Wise Error Rate), 即 m 次检验中出现任何 type-I error 的概率, 为 α (保守)
- ② 控制 FDR (False Discovery Rate), 即 (允许出现 type-I error) type-I error 占显著结果的比例, 为 α (宽松)
- ③ 利用 subhypotheses 的结果 (比如 p 值) 构造新的 global testing, 确保新的 testing 的 type-I error 为 α (仅适用于 global testing)

以下是一些 multiple hypothesis testing 方法的汇总:

一、控制Family-Wise Error Rate (FWER) 的方法

目标: 确保所有检验中至少出现一个假阳性的概率 $\leq \alpha$ (严格控假阳性, 适合保守场景)。

方法	原理	适用场景	优缺点
Bonferroni 校正	将显著性水平调整为 α/m (m 为检验次数)	小规模检验 (如10次以下), 或需要严格控FWER的验证性研究 (如临床试验)	优点: 简单易用。 缺点: 过于保守, 大样本易漏真效应。
Šidák校正	调整显著性水平为 $1 - (1 - \alpha)^{(1/m)}$	类似Bonferroni, 但假设检验独立时更高效	优点: 比Bonferroni略高效。 缺点: 仍需独立假设, 仍保守。
Holm-Bonferroni	逐步法: 按 p 值排序, 依次比较 $\alpha/(m - k + 1)$ (k 为当前排名)	中等规模检验, 比Bonferroni更高效 (如心理学实验中的多重比较)	优点: 比Bonferroni更敏感。 缺点: 仍保守, 不适用于极大规模检验。
Hochberg 法	逐步法: 按 p 值排序, 依次比较 $\alpha/(m - k + 1)$, 但允许提前拒绝假设	类似Holm, 但更高效 (需检验间独立或正相关)	优点: 比Holm更敏感。 缺点: 对相关性敏感, 需满足特定条件。
Dunnett法	专为多个实验组与一个对照组比较设计 (如药物剂量试验)	多组与单一对照组比较 (如A vs Control, B vs Control, C vs Control)	优点: 针对设计优化, 高效。 缺点: 仅适用于特定比较类型。
Tukey HSD	基于学生化极差分布, 控制所有两两比较的 FWER	ANOVA后的事后两两比较 (如比较A/B/C三组的均值差异)	优点: 精确控制FWER。 缺点: 仅适用于平衡设计, 计算复杂。

二、控制False Discovery Rate (FDR) 的方法

目标: 控制被拒绝的假设中假阳性的比例 $\leq \alpha$ (适合大规模探索性研究)。

方法	原理	适用场景	优缺点
Benjamini-Hochberg (BH)	动态调整阈值：对排序后的p值，比较 $p \leq (k/m)\alpha$ (k为排名)	大规模检验（如基因组学、神经影像学中的数万次检验）	优点：平衡敏感性与特异性。 缺点：需检验独立或正相关。
Benjamini-Yekutieli (BY)	调整BH方法以适用于任意相关性结构（阈值改为 $(k/m)\alpha/\Sigma(1/i)$ ）	检验间存在未知或复杂相关性（如经济学中的多重回归分析）	优点：稳健性强。 缺点：更保守，敏感性降低。
Storey's q-value	基于估计的 π_0 （真实零假设比例），计算FDR的估计值而非严格控制	超大规模数据（如全基因组关联分析GWAS），允许灵活权衡敏感性与特异性	优点：高效利用数据信息。 缺点：依赖 π_0 估计的准确性。

三、全局假设检验（Global Hypothesis Testing）方法

目标：检验所有零假设是否同时成立（如“所有基因均无差异”）。

方法	原理	适用场景	优缺点
Fisher's Combination Test	将多个p值通过统计量 $T = -2 \sum \ln(p_i)$ 合并，服从卡方分布	小到中等规模检验，需综合多个独立检验结果（如元分析中的效应整合）	优点：对微弱效应敏感。 缺点：需独立假设，对异质性敏感。
Simes Test	若至少有一个p值 $\leq k\alpha/m$ (k为排名)，则拒绝全局零假设	初步筛选是否存在任何效应（如药物开发中的初步筛选）	优点：简单易用。 缺点：需检验间独立或正相关。
Hotelling's T²	多元均值检验（扩展t检验到多维）	多变量均值向量的整体检验（如同时检验身高、体重、血压的组间差异）	优点：考虑变量相关性。 缺点：需正态分布和大样本。

四、其他特殊场景方法

方法	原理	适用场景	控制目标
Permutation Test	通过重抽样生成零分布，计算经验p值	非参数检验、复杂设计或检验分布未知时（如生态学中的多样性分析）	FWER或FDR均可控制
Resampling-based FDR	利用Bootstrap或置换法估计FDR	高维数据且检验间存在复杂依赖关系（如脑网络分析）	FDR

总结

- FWER方法**：严格控假阳性，适合“不容有错”的场景（如药物审批）。
- FDR方法**：允许可控的假阳性比例，适合“大海捞针”式研究（如生物标志物发现）。
- 全局检验**：快速判断是否存在任何效应，避免无意义的多重比较。

根据你的数据规模、研究目标和错误容忍度，选择最适合的方法！

在接下来的讨论中，我们将主要研究 *global testing / simultaneous multiple hypothesis testing* 并介绍一部分 *multiple hypothesis testing* 方法

§1 Global testing problem 的形式

1. Definition: Global testing problem (Simultaneous multiple hypothesis testing problem)

Global testing problem (Simultaneous multiple hypothesis testing problem) 的形式为:

$$H_0 = \bigcap_{i=1}^m H_{0i} \quad v.s. \quad H_1 = \bigcup_{i=1}^m H_{1i} \quad (\text{研究 } m \text{ 个 individual null 是否同时成立})$$

其中,

- ① fix sample size n 后, m 为 non-infinite number
- ② sample size n 允许 goes to infinity
- ③ 允许 $m = m(n)$, 即 m 取决于 sample size n

注: 此处我们不要求 m 为 low-dimensional ($\rightarrow 0$ as $n \rightarrow \infty$), 或 high-dimensional ($\rightarrow \infty$ as $n \rightarrow \infty$), 或 ultra-dimensional ($\log m \rightarrow \infty$ as $n \rightarrow \infty$).

e.g. 一个生物统计中的例子:

In the global testing problem, the global null we are interested states that all of the individual nulls are true. One may often encounter such problems in bioinformatics field. For example, suppose we have n genes and data about expression levels for each gene among healthy individuals and those with breast cancer,

Expression level of Gene $i \in \{1, \dots, n\}$

k Healthy patients: $x_{i1}^{(0)}, x_{i2}^{(0)}, \dots, x_{ik}^{(0)}$

ℓ breast cancer patients: $x_{i1}^{(1)}, x_{i2}^{(1)}, \dots, x_{i\ell}^{(1)}$

Naturally, the i -th null hypothesis, denoted as H_{0i} , states that the mean expression level of the i -th gene is the same in both groups of patients, i.e.,

$$H_{0i} : \mathbb{E}(x_{ij}^{(0)}) = \mathbb{E}(x_{ij}^{(1)}) \quad v.s. \quad H_{1i} : \mathbb{E}(x_{ij}^{(0)}) \neq \mathbb{E}(x_{ij}^{(1)}). \quad (\text{基因 } i \text{ 是否差异表达})$$

For simplicity, we assume for each hypothesis H_{0i} v.s. H_{1i} , we already have a test statistic T_i associated with a rejection region $R_i(\alpha') = \{T_i > c(\alpha')\}$, hence, a p-value p_i that can be defined as

$$p_i = \mathbb{P}(T_i \geq t_i | H_0) = 1 - F_{T_i}(t_i)$$

where t_i is the observed value of the statistic T_i . When the distribution function F_{T_i} is continuous under the null hypothesis, we have $F_{T_i}(T_i) \sim \text{Uniform}[0, 1]$, and hence the $p_i \sim \text{Uniform}[0, 1]$. But in general, when we have rather irregular rejection region or when the distribution function F_{T_i} is discontinuous under the null hypothesis, the distribution of p_i can still be hard to track.

注: 为了避免麻烦, 接下来的讨论中默认 $p_i \sim \text{Uniform}(0, 1)$ for $i = 1, \dots, m$

§2 Global testing problem 的方法: 控制 significance level 的方法 (控制 FWER)

1. Definition: Bonferroni approach

令 global testing problem 的 significance level 为 α

则 Bonferroni approach reject H_0 , 若

$$\min_{1 \leq i \leq m} p_i < \frac{\alpha}{m}$$

注: ① Bonferroni approach 的 intuition 是: 先检验每个 subhypothesis at level $\frac{\alpha}{m}$, 若有任何 H_{0i} 被拒绝, 则拒绝 global null H_0

②* Bonferroni approach 也适用于 pairwise multiple hypothesis testing, 只需把 levels 调整为 $\frac{\alpha}{m}$

③ Bonferroni approach 对 $\{H_{0i}\}$ 没有任何要求, 永远可以使用它得到一个结果

④ Bonferroni approach 非常严格, 只有当某个 p-value 非常 significant 时才会 reject H_0 . 这可能导致 power 较小. 若我们只希望找出极个别非常 significant 的 genes, 可以考虑使用.

证明: Bonferroni approach 的 validity (Type-I error $\leq \alpha$)

$$\begin{aligned}\text{Type-I error} &= P_{H_0}(\bigcup_{i=1}^m \{P_i < \frac{\alpha}{m}\}) \\ &\leq \sum_{i=1}^m P_{H_0}(P_i < \frac{\alpha}{m}) \\ &= \sum_{i=1}^m \frac{\alpha}{m} \\ &= \alpha\end{aligned}$$

注: 即便证明中的放缩非常宽松, 但至少对于 independent H_{0i} , Bonferroni approach 的 Type-I error 会非常接近 α :

$$\begin{aligned}\text{Type-I error} &= 1 - P_{H_0}(\bigcap_{i=1}^m \{P_i \geq \frac{\alpha}{m}\}) \\ &= 1 - (1 - \frac{\alpha}{m})^m && (P_i \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0,1) \text{ under } H_0) \\ &\rightarrow 1 - e^{-\alpha} && (\text{as } m \rightarrow \infty) \\ &\approx 1 - (1 - \alpha + \frac{1}{2}e^{-\tilde{\alpha}}\alpha^2) && (\text{Taylor's expansion, } \tilde{\alpha} \in (0, \alpha)) \\ &\approx \alpha && (\text{for small } \alpha)\end{aligned}$$

2. Definition: Positive / Negative quadrant dependent

令 ① $\{P_i\}_{1 \leq i \leq m}$ 为一个 sequence of r.v.

② $\{P_i\}_{1 \leq i \leq m}$ 的 distribution function 为 continuous

则 ① $\{P_i\}_{1 \leq i \leq m}$ 为 positive quadrant dependent, 若对 $\forall \mu_1, \dots, \mu_m \in \mathbb{R}$, 有

$$PQD(\mu_1, \dots, \mu_m) := P(\bigcap_{i=1}^m \{P_i \leq \mu_i\}) - \prod_{i=1}^m P(P_i \leq \mu_i) \geq 0$$

或

$$\widetilde{PQD}(\mu_1, \dots, \mu_m) := P(\bigcap_{i=1}^m \{P_i \geq \mu_i\}) - \prod_{i=1}^m P(P_i \geq \mu_i) \geq 0$$

② $\{P_i\}_{1 \leq i \leq m}$ 为 negative quadrant dependent, 若对 $\forall \mu_1, \dots, \mu_m \in \mathbb{R}$, 有

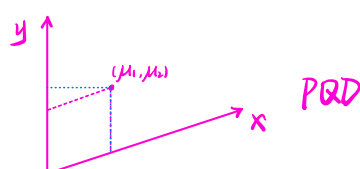
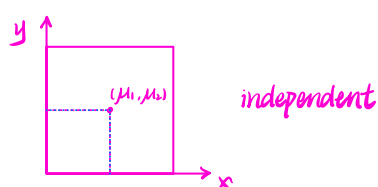
$$PQD(\mu_1, \dots, \mu_m) := P(\bigcap_{i=1}^m \{P_i \leq \mu_i\}) - \prod_{i=1}^m P(P_i \leq \mu_i) \leq 0$$

或

$$\widetilde{PQD}(\mu_1, \dots, \mu_m) := P(\bigcap_{i=1}^m \{P_i \geq \mu_i\}) - \prod_{i=1}^m P(P_i \geq \mu_i) \leq 0$$

注: 若 x 和 y 为 NQD, 则 x 取值较小时, y 更可能取较大的值.

若 x 和 y 为 PQD, 则 x 取值较小时, y 更可能取较小的值.



3. Definition: Šidák approach

令 ① global testing problem 的 significance level 为 α

② subtests, BP $\{P_i\}_{1 \leq i \leq m}$, 为 mutually independent 或 positive quadrant dependent

则 Šidák approach reject H_0 , 若

$$\min_{1 \leq i \leq m} P_i < 1 - (1 - \alpha)^{\frac{1}{m}}$$

注: ① Šidák approach 的 intuition 是: 先检验每个 subhypothesis at level $\alpha' = 1 - (1 - \alpha)^{\frac{1}{m}}$, 若有任何 H_{0i} 被拒绝, 则拒绝 global null H_0

②* Šidák approach 也适用于 pairwise multiple hypothesis testing, 只需把 levels 调整为 $1 - (1 - \alpha)^{\frac{1}{m}}$

③ Šidák approach 对 $\{H_{0i}\}$ 有要求, 但可以控制 Type-I error exactly 等于 α

④ Šidák approach 的构造非常 intuitive:

若 $\{P_i\}_{1 \leq i \leq m}$ mutually independent, 则我们希望控制

$$\begin{aligned} \text{Type-I error} &= 1 - P_{H_0}(\bigcap_{i=1}^m \{P_i \geq \frac{\alpha'}{m}\}) \\ &= 1 - (1 - \frac{\alpha'}{m})^m \\ &\triangleq \alpha \end{aligned}$$

$$\Rightarrow \text{choose } \alpha' = 1 - (1 - \alpha)^{\frac{1}{m}}$$

证明: Šidák approach 的 validity (Type-I error $\leq \alpha$)

$$\begin{aligned} \text{Type-I error} &= P_{H_0}(\bigcup_{i=1}^m \{P_i < 1 - (1 - \alpha)^{\frac{1}{m}}\}) \\ &= 1 - P_{H_0}(\bigcap_{i=1}^m \{P_i \geq 1 - (1 - \alpha)^{\frac{1}{m}}\}) \\ &\leq 1 - \prod_{i=1}^m P_{H_0}(P_i \geq 1 - (1 - \alpha)^{\frac{1}{m}}) \\ &= 1 - \prod_{i=1}^m (1 - \alpha)^{\frac{1}{m}} \\ &= \alpha \end{aligned}$$

§3 Global testing problem 的方法: combination type test (构造新的 global testing)

1. Definition: Fisher's combination test

令 ① global testing problem 的 significance level 为 α

② subtests, BP $\{P_i\}_{1 \leq i \leq m}$, 为 mutually independent

则 Fisher's combination test reject H_0 , 若

$$T = -\sum_{i=1}^m 2 \cdot \log P_i > \chi_{2m}^2(1 - \alpha)$$

注: ① 若 p-values 越小, T 越大, test 越倾向于 reject H_0

② 区别于 Bonferroni approach, Fisher's combination test 适用于 dense data, 即有很多 small effects 而不是个别 strong effects 的情况.

2. Proposition: Fisher's combination test statistic 在 H_0 下的分布 (Fisher's combination test 的 validity)

若 ① subtests, 即 $\{P_i\}_{1 \leq i \leq m}$, 为 mutually independent

② $\{P_i\}_{1 \leq i \leq m}$ 在 H_0 下服从 uniform distribution

则在 H_0 下,

$$T \sim \chi^2_{2m}$$

证明:

由于 H_0 下 $P_i \sim \text{Uniform}(0, 1)$, 且 $\text{Exp}(2)$ 的 distribution function 为

$$F(x) = e^{-x/2}$$

因此根据 Inverse transformation method, 有

$$F^{-1}(P_i) = -2 \log P_i \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(2)$$

因此,

$$T = -\sum_{i=1}^m 2 \cdot \log P_i \sim \text{Gamma}(m, 2) \stackrel{d}{=} \chi^2_{2m}$$

3. Definition: Simes's approach

令 ① global testing problem 的 significance level 为 α

② subtests, 即 $\{P_i\}_{1 \leq i \leq m}$, 为 mutually independent

③ $P_{(1)} \leq P_{(2)} \leq \dots \leq P_{(m)}$ 为 $\{P_i\}_{1 \leq i \leq m}$ 的 order statistics

则 Simes's approach reject H_0 , 若

$$T_{\text{Simes}} = \min_{1 \leq i \leq m} \{P_{(i)} \cdot \frac{m}{i}\}$$

注: 若将每个 subtest 的 p 值 $P_{(i)}$ 调整为 $P_{(i)} \cdot \frac{m}{i}$, 则变为 Benjamini-Hochberg correction, 一种控制 FDR 的方法

4. Proposition: Simes's approach test statistic 在 H_0 下的分布 (Simes's approach 的 validity)

若 ① subtests, 即 $\{P_i\}_{1 \leq i \leq m}$, 为 mutually independent

② $\{P_i\}_{1 \leq i \leq m}$ 在 H_0 下服从 uniform distribution

则在 H_0 下,

$$T_{\text{Simes}} \sim \text{Uniform}(0, 1)$$

§4 Global testing problem 的方法: 用于特定形式的 hypothesis testing 问题 (控制 FWER, 但只用于特殊问题)

1. Definition: (Post-hoc) multiple comparison problem

(Post-hoc) multiple comparison problem 的每个 H_{0ij} 为一个 comparison problem:

$$H_{0ij}: \mu_i = \mu_j \quad \text{v.s.} \quad H_{1ij}: \mu_i \neq \mu_j, \quad i, j = 1, 2, \dots, m, \quad i \neq j$$

其可被进一步分为:

① Pairwise multiple comparison problem: 分别用 significant level α 检验每个

$$H_{0ij} \quad \text{v.s.} \quad H_{1ij}$$

② Simultaneous multiple comparison problem: 用一个 significant level α 检验

$$H_0 = \bigcap \{H_{0ij} : \mu_i = \mu_j\} \quad \text{v.s.} \quad H_1 = \bigcup \{H_{1ij} : \mu_i \neq \mu_j\} \quad , \quad i, j = 1, 2, \dots, m, \quad i \neq j$$

注: ① multiple comparison problem 是 multiple hypothesis testing 的特例

② Simultaneous multiple comparison problem 的 hypothesis 等价于:

$$H_0: \mu_1 = \dots = \mu_m \quad \text{v.s.} \quad H_1: \mu_i \neq \mu_j \text{ for some } i, j \text{ where } i \neq j$$

最常见的 multiple comparison problem 是 post-hoc ANOVA

2. Definition: post-hoc ANOVA

令 ① m 表示 group 的数量

② n_i 表示 i -th group 的 data 数

③ $N = \sum_{i=1}^m n_i$ 表示总 data 数

④ μ_i 表示 i -th group 的 mean

⑤ $\{Y_{ij}, j=1, \dots, n_i, i=1, \dots, m\}$ generated from homoscedastic normal distribution:

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij} \quad , \quad j=1, \dots, n_i, \quad i=1, \dots, m$$

其中,

$$\varepsilon_{ij} \stackrel{\text{iid.}}{\sim} N(0, \sigma^2)$$

则 ① post-hoc ANOVA 关注 μ_i 间是否有差异

② group mean 被定义为:

$$\bar{Y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij}$$

③ mean square error 被定义为:

$$S_E^2 = \frac{1}{N-m} \sum_{i=1}^m \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2, \quad i=1, \dots, m$$

注: 对于 post-hoc ANOVA setting 和不同的 comparison problem, 有多种解决方案:

① 用于 Pairwise multiple comparison problem: SNK approach

② 用于 Simultaneous multiple comparison problem: Dunnett's approach

③ 兼容两种 problems: Fisher's LSD (least significant difference) with Bonferroni's approach

3. Definition: Dunnett's approach 的 hypothesis

Dunnett's approach 适用于 multiple comparison problem, 其 hypothesis 为

$$H_0 = \bigcap_{j=2}^m \{H_{0j} : \mu_1 = \mu_j\} \quad \text{v.s.} \quad H_1 = \bigcup_{j=2}^m \{H_{1j} : \mu_1 \neq \mu_j\}$$

注: 区别于检验所有 pairwise difference, Dunnett's approach 主要比较 treatment groups (group 2, ..., m) 和 control group (group 1)

4. Definition: Dunnett's approach

令 Simultaneous multiple comparison problem 的 significance level 为 α

则 ① Dunnett's approach 的 test statistic 为:

where $\epsilon_{ij} \sim_{i.i.d} N(0, \sigma^2)$, i.e., the data are generated from m groups with $N = \sum_{i=1}^m n_i$. Denote

$$\text{Group mean: } \bar{Y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij}, \quad i = 1, \dots, m,$$

$$\text{Mean square error: } S_E^2 = \frac{1}{N - m} \sum_{i=1}^m \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2, \quad i = 1, \dots, m,$$

In the following content, we will introduce several approaches that are specifically designed for post-hoc multiple comparison problems. But as a preview, we emphasize that those post-hoc multiple comparison procedures are difficult to compare as some of them are designed for pairwise comparison (E.g., one Step-down procedure called SNK approach), some of them are designed for simultaneous comparison (E.g., Dunnett's approach), while some of them are designed for pairwise comparison but are adaptive to simultaneous comparison if we combine them with approaches introduced in Section 2 (E.g., combine Fisher's LSD with Bonferroni's approach).

3.1. Approaches for simultaneous multiple comparison problem

We introduce some approaches that are designed for simultaneous multiple comparison problem (3.2). Apparently, they are not as general as approaches introduced in Section 2 as they only focus on (3.2) though they share similar ideas:

- ♣ **Dunnett's Approach:**

Unlike taking intersection with respect to both i and j in the H_0 of (3.2), Dunnett's approach is usually invoked for an slightly different simultaneous multiple comparison problem:

$$H_0 = \bigcap_{j=2}^m \{H_{0j} : \mu_1 = \mu_j\} \quad v.s. \quad H_1 = \bigcup_{j=2}^m \{H_{1j} : \mu_1 \neq \mu_j\},$$

for some $j = 2, \dots, m$. (3.4)

In other words, instead of interested in testing all pairwise differences, Dunnett's approach focus on the comparison between other groups (group 2, 3, $\dots$, m , one may think them as various treatment groups) to a specific group (denoted as group 1, and one may think it as the control group). This restriction can of course been extend to (3.2) but with some complicity in computing the corresponding critical value (investigate corresponding limiting distribution).

Procedure: Given a significant level α for the simultaneous multiple comparison problem (3.4), for $j = 2, \dots, m$, define the test statistic

$$T_j \triangleq \frac{\bar{Y}_j - \bar{Y}_1}{S_E \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_j}}}, \quad \text{and } T = \max_{2 \leq j \leq m} |T_j|$$

In Dunnett's original paper [Dunnett \(1955\)](#), the procedures then follows by try to find the vector of critical values $(d_2, \dots, d_m)$, such that

$$\mathbb{P}\left(|T_j| \leq d_j, j = 2, \dots, m\right) = 1 - \alpha$$

and we reject the global null hypothesis H_0 in (3.4) whenever

$$\max_{2 \leq j \leq m} (|T_j| - d_j) > 0$$

Remark 3.1. Generally, the critical value $(d_2, \dots, d_m)$ is hard to find, but [Dunnett and Sobel \(1955\)](#) gives the following proposition,

Proposition 3.2. Assume the data is generated from homoscedastic normal distribution (3.3), then statistics $\{T_j\}_{2 \leq j \leq m}$ are positively quadrant dependent.

Therefore we may pick a critical value slightly conservative, i.e., $d_j = t_{N-m}(1 - \alpha/2)$, since

$$\prod_{j=2}^m \mathbb{P}\left(|T_j| \leq d_j\right) \leq \mathbb{P}\left(|T_j| \leq d_j, j = 2, \dots, m\right) = 1 - \alpha,$$

and each T_j follows t -distribution with degree of freedom $N - m$.

So Dunnett's approach is valid.

4. Optimality of Bonferroni's Global Test

Despite being a simple procedure, we show here that Bonferroni's method is somehow optimal for testing against sparse alternatives. This claim relies on power calculations, which require us to specify alternatives.

Consider an independent Gaussian sequence model:

$$Y_i \sim \mathcal{N}(\mu_i, 1), \quad i = 1, \dots, n,$$

where $Y = \{Y_1, \dots, Y_n\}$ are mutually independent with each other. We are interested in the n hypotheses

$$H_{0i} : \mu_i = 0 \text{ v.s. } H_{1i} : \mu_i \neq 0,$$

so that in this case, the global null asserts that all the means μ_i vanish, while under the alternative H_1 , at least some means $\mu_i \neq 0$.

When use Bonferroni's approach, we reject the overall H_0 if

$$\max_{1 \leq i \leq n} |Y_i| > \mu(1 - \alpha/(2n))$$

with $\mu(1 - \alpha/(2n))$ being the $(1 - \alpha/(2n))$ -th quantile of the standard normal distribution. Put another way, Bonferroni rejects the global null hypothesis if the largest Y_i is large enough. For the special case where the n tests are mutually independent, we may calculate that

$$\mathbb{P}_{H_0}(\text{Type I Error}) := q(\alpha) \approx 1 - e^{-\alpha} \approx \alpha.$$

To see this, notice that according to Markov's inequality, for $Z \sim N(0, 1)$, we have

$$\begin{aligned} \mathbb{P}(Z > t) &= \int_t^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \frac{e^{-\frac{t^2}{2}}}{\sqrt{2\pi}} \cdot \int_0^\infty e^{-tz - z^2/2} dz \\ &\leq \frac{e^{-\frac{t^2}{2}}}{\sqrt{2\pi}} \cdot \int_0^\infty e^{-tz} dz = \frac{e^{-\frac{t^2}{2}}}{t \cdot \sqrt{2\pi}} = \frac{\phi(t)}{t} \end{aligned}$$

where $\phi(\cdot)$ is the density function of the standard normal distribution. On the other hand, we have

$$\begin{aligned} \mathbb{P}(Z > t) &= \int_t^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \geq \int_t^\infty (1 - 3x^{-4}) \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &= \left(\frac{1}{t} - \frac{1}{t^3} \right) \frac{e^{-\frac{t^2}{2}}}{\sqrt{2\pi}} = \left(\frac{1}{t} - \frac{1}{t^3} \right) \phi(t). \end{aligned}$$

Hence, overall we have for $\forall t > 0$,

$$\frac{\phi(t)}{t} \left(1 - \frac{1}{t^2} \right) \leq \mathbb{P}(Z > t) \leq \frac{\phi(t)}{t},$$

That is, for large t , $\phi(t)/t$ is a good approximation to the normal tail probability and we may use it to calculate t , which is the Gaussian quantile. Roughly speaking, for large n

$$\mathbb{P}(Z > t) = \alpha/n \quad \Leftrightarrow \quad \frac{\phi(t)}{t} \approx \alpha/n.$$

Holding α fixed, then, we can show that for large n ,

$$|\mu(1 - \alpha/(2n))| \approx \sqrt{2 \log(2n)} \left[1 - \frac{1}{4} \frac{\log \log(2n)}{\log(2n)} \right] \approx \sqrt{2 \log(2n)}.$$

Hence, the quantiles grow like $\sqrt{2 \log(2n)}$, with a small correction factor. Figure 1 plots $\mu(1 - \alpha/n)$ and $\sqrt{2 \log n}$. Notice that Bonferroni then basically amounts to rejecting when $\max |Y_i| > \sqrt{2 \log n}$.

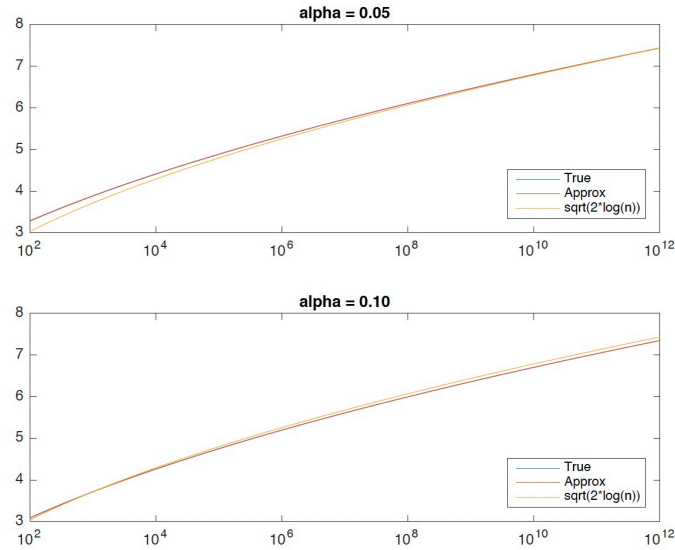


Figure 1: Bonferroni Approach

One remarkable fact about all of this is that there is (asymptotically) no dependence on α . That is, whatever α we use, our rejection threshold for $\max |Y_i|$ is asymptotic to $\sqrt{2 \log n}$. This is a consequence of the fact that, under H_0 ,

$$\frac{\max |y_i|}{\sqrt{2 \log n}} \xrightarrow{p} 1.$$

In other words, the first order term $\sqrt{2 \log n}$ asymptotically dominates the terms containing α . If we set

$$B = 2 \log(n/\alpha) - \log(2\pi) = 2 \log(n/\alpha) - 1.8379.$$

Then

$$|\mu(1 - \alpha/n)| \approx \sqrt{B \left(1 - \frac{\log B}{B}\right)}.$$

Figure 1 shows that this approximation is nearly indistinguishable from $\mu(1 - \alpha/n)$, even for modest values of n .

5. Sharp Detection Threshold for the "Needle in a Haystack"

Definition 5.1 (Asymptotic Power). For a sequence of problems with $n \rightarrow \infty$, the asymptotic power is the limiting power

$$\mathbb{P}_{H_1}(\text{Reject } H_0).$$

1. The first question is: how powerful is Bonferroni?
 2. The second question is: what is the limiting power $\mathbb{P}_{H_1}(\max Y_i > \mu(1 - \alpha/(2n)))$?

A natural question arise that we want to see how powerful is Bonferroni, or, put it another way, what is the limiting power $\mathbb{P}_{H_1}(\max Y_i > \mu(1 - \alpha/(2n)))$?

To answer the above question, we first look at a simple problem called the **Needle in a Haystack Problem**, for which we specify the alternative hypotheses. The needle in a haystack problem is this: under the alternative, we have one and only one $\mu_i = \mu(n) > 0$. But we don't know which one. For this problem, we shall see that the answer to the power question depends very sensitively on the limiting ratio

$$\frac{\mu(n)}{\sqrt{2 \log n}},$$

where the n in $\mu(n)$ are emphasizing the dependence between this non-zero mean and the sample size. There are two cases:

- (i) Asymptotic full power above threshold: Suppose $\mu(n) > (1 + \epsilon)\sqrt{2 \log n}$. Then, without loss of generality, assume that $\mu_1 = \mu(n)$, we have

$$\begin{aligned}
 \mathbb{P}_{H_1}(\text{Reject } H_0) &= \mathbb{P}_{H_1}(\max |Y_i| > \mu(1 - \alpha/(2n))) \\
 &\geq \mathbb{P}_{H_1}(Y_1 > \mu(1 - \alpha/(2n))) \\
 &= \Phi(\mu(1 - \alpha/(2n)) - \mu(n)) \rightarrow 1.
 \end{aligned}$$

- (ii) Asymptotic powerlessness below threshold: Suppose $\mu(n) < (1 - \epsilon)\sqrt{2 \log n}$. Then, without loss of generality, assume that $\mu_1 = \mu(n)$, we have

$$\begin{aligned}
 \mathbb{P}_{H_1}(\text{Reject } H_0) &= \mathbb{P}_{H_1}(\max |Y_i| > \mu(1 - \alpha/(2n))) \\
 &\leq \mathbb{P}_{H_1}(Y_1 > \mu(1 - \alpha/(2n))) + \mathbb{P}\left(\max_{i>1} |Y_i| > \mu(1 - \alpha/(2n))\right) \\
 &= \Phi(\mu(1 - \alpha/(2n)) - \mu(n)) + \mathbb{P}\left(\max_{i>1} |Y_i| > \mu(1 - \alpha/(2n))\right) \\
 &\rightarrow 0 + \alpha = \alpha.
 \end{aligned}$$

meaning this is a bad test because we can obtain the same level and power by flipping a biased coin that rejects α of the time.

Therefore, we may effectively see that $\sqrt{2 \log n}$ constitutes a sharp detection threshold. When $\mu(n) = (1 + \epsilon)\sqrt{2 \log n}$, we can always detect the needle that $\mu_1 > 0$. We can even achieve

$$\mathbb{P}_{H_0}(\text{Type I Error}) \rightarrow 0 \quad \text{and} \quad \mathbb{P}_{H_1}(\text{Type II Error}) \rightarrow 0.$$

if we use $\sqrt{2 \log n}$ instead of $\mu(1 - \alpha/(2n))$ as our threshold. In other words, asymptotically we make no mistakes.

However, when $\mu(n) = (1 - \epsilon)\sqrt{2 \log n}$, with $q(\alpha) = 1 - e^{-\alpha} \approx \alpha$ being the asymptotic size, Bonferroni's global test gives

$$\mathbb{P}_{H_0}(\text{Type I Error}) \rightarrow q(\alpha) \quad \text{and} \quad \mathbb{P}_{H_1}(\text{Type II Error}) \rightarrow 1 - q(\alpha).$$

that is, it does no better than flipping a coin.

References

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